



Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 5-Solutions

1. Given the potential:

$$V = \frac{10}{r^2} \sin \theta \cos \phi$$

a. Find the electric flux density $\mathbf{D}$ at $(2, \pi/2, 0)$.

b. Calculate the work done in moving a $10\text{-}\mu\text{C}$ charge from point $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$.

Solution

a. $\mathbf{D} = \epsilon_0 \mathbf{E}$

But

$$\mathbf{E} = -\nabla V = -\left[\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi\right]$$

$$\mathbf{E} = \frac{20}{r^3} \sin \theta \cos \phi \mathbf{a}_r - \frac{10}{r^3} \cos \theta \cos \phi \mathbf{a}_\theta + \frac{10}{r^3} \sin \phi \mathbf{a}_\phi$$

At $(2, \pi/2, 0)$:

$$\mathbf{D} = \epsilon_0 \mathbf{E} \left(r = 2, \theta = \frac{\pi}{2}, \phi = 0 \right) = \epsilon_0 \left(\frac{20}{8} \mathbf{a}_r - 0 \mathbf{a}_\theta + 0 \mathbf{a}_\phi \right)$$

$$\mathbf{D} = 2.5 \epsilon_0 \mathbf{a}_r \text{ C/m}^2 = 22.1 \mathbf{a}_r \text{ pC/m}^2$$

b. The work done can be found in two ways, using either $\mathbf{E}$ or V .

Method 1:

$$W = -Q \int \mathbf{E} \cdot d\mathbf{l}$$

or

$$-\frac{W}{Q} = \int \mathbf{E} \cdot d\mathbf{l}$$

and because the electrostatic field is conservative, the integration path is immaterial. Hence, the work done in moving Q from $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$ is the same as that in moving Q from A to $\hat{A}$, from $\hat{A}$ to $\hat{B}$, and from $\hat{B}$ to B , where:

$$A(1, 30^\circ, 120^\circ) \xrightarrow{dl=dr\mathbf{a}_r} \hat{A}(4, 30^\circ, 120^\circ) \xrightarrow{dl=r d\theta\mathbf{a}_\theta} \hat{B}(4, 90^\circ, 60^\circ) \xrightarrow{dl=r \sin\theta d\phi\mathbf{a}_\phi} B(4, 90^\circ, 60^\circ)$$

That is, instead of moving Q directly from A and B , it is moved from $A \rightarrow \hat{A}$, $\hat{A} \rightarrow \hat{B}$, $\hat{B} \rightarrow B$ so that only one variable is changed at a time. This makes the line integral a lot easier to evaluate. Thus:

$$\begin{aligned} -\frac{W}{Q} &= -\frac{1}{Q} (W_{A\hat{A}} + W_{\hat{A}\hat{B}} + W_{\hat{B}B}) \\ -\frac{W}{Q} &= \left(\int_{A\hat{A}} + \int_{\hat{A}\hat{B}} + \int_{\hat{B}B} \right) \mathbf{E} \cdot d\mathbf{l} \\ -\frac{W}{Q} &= \int_{r=1}^4 \frac{20 \sin\theta \cos\phi}{r^3} dr \Big|_{\theta=30^\circ, \phi=120^\circ} + \int_{\theta=30^\circ}^{90^\circ} \frac{-10 \cos\theta \cos\phi}{r^3} r d\theta \Big|_{r=4, \phi=120^\circ} \\ &\quad + \int_{\phi=120^\circ}^{60^\circ} \frac{10 \sin\phi}{r^3} r \sin\theta d\phi \Big|_{r=4, \theta=90^\circ} \\ -\frac{W}{Q} &= 20 \left(\frac{1}{2} \right) \left(\frac{-1}{2} \right) \left[-\frac{1}{2r^2} \right]_{r=1}^4 - \frac{10(-1)}{16} \frac{1}{2} \sin\theta \Big|_{30^\circ}^{90^\circ} + \frac{10}{16} (1) [-\cos\phi]_{120^\circ}^{60^\circ} \\ -\frac{W}{Q} &= -\frac{75}{32} + \frac{5}{32} - \frac{10}{16} \end{aligned}$$

or

$$W = \frac{45}{16} Q = 28.125 \mu J$$

Method 2:

Since V is known, this method is a lot easier.

$$W = -Q \int_A^B \mathbf{E} \cdot d\mathbf{l} = QV_{AB}$$

$$W = Q(V_B - V_A) = 10 \left(\frac{10}{16} \sin 90^\circ \cos 60^\circ - \frac{10}{1} \sin 30^\circ \cos 120^\circ \right) \cdot 10^{-6}$$

$$W = 10 \left(\frac{10}{32} - \frac{-5}{2} \right) \cdot 10^{-6} = 28.125 \mu J$$

As obtained before.

2. Two dipoles with dipole moments $-5\mathbf{a}_z \text{ nC/m}$ and $9\mathbf{a}_z \text{ nC/m}$ are located at points $(0, 0, -2)$ and $(0, 0, 3)$, respectively. Find the potential at the origin.

Solution

$$V = \sum_{k=1}^2 \frac{p_k \cdot r_k}{4\pi\epsilon_o r_k^3} = \frac{1}{4\pi\epsilon_o} \left[\frac{p_1 \cdot r_1}{r_1^3} + \frac{p_2 \cdot r_2}{r_2^3} \right]$$

where:

$$p_1 = -5\mathbf{a}_z \quad r_1 = (0, 0, 0) - (0, 0, -2) = 2\mathbf{a}_z \quad r_1 = |r_1| = 2$$

$$p_2 = 9\mathbf{a}_z \quad r_2 = (0, 0, 0) - (0, 0, 3) = -3\mathbf{a}_z \quad r_2 = |r_2| = 3$$

hence:

$$V = \frac{1}{4\pi \frac{10^{-9}}{36\pi}} \left[\frac{-10}{2^3} - \frac{27}{3^3} \right] \cdot 10^{-9} = -20.25 \text{ V}$$

3. Three point charges -1 nC , 4 nC , and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, $(1, 0, 0)$, respectively. Find the energy in the system.

Solution

$$W = W_1 + W_2 + W_3 = 0 + Q_2 V_{21} + Q_3 (V_{31} + V_{32})$$

$$W = Q_2 \cdot \frac{Q_1}{4\pi\epsilon_o |(0,0,1) - (0,0,0)|} + \frac{Q_3}{4\pi\epsilon_o} \left[\frac{Q_1}{|(1,0,0) - (0,0,0)|} + \frac{Q_2}{|(1,0,0) - (0,0,1)|} \right]$$

$$W = \frac{1}{4\pi\epsilon_o} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right)$$

$$W = \frac{1}{4\pi \frac{10^{-9}}{36\pi}} \left(-4 - 3 + \frac{12}{\sqrt{2}} \right) \cdot 10^{-18}$$

$$W = 9 \left(\frac{12}{\sqrt{2}} - 7 \right) \text{ nJ} = 13.37 \text{ nJ}$$

Alternatively:

$$W = \frac{1}{2} \sum_{k=1}^3 Q_k V_k = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3)$$

$$W = \frac{Q_1}{2} \left[\frac{Q_2}{4\pi\epsilon_o(1)} + \frac{Q_3}{4\pi\epsilon_o(1)} \right] + \frac{Q_2}{2} \left[\frac{Q_1}{4\pi\epsilon_o(1)} + \frac{Q_3}{4\pi\epsilon_o(\sqrt{2})} \right] + \frac{Q_3}{2} \left[\frac{Q_1}{4\pi\epsilon_o(1)} + \frac{Q_2}{4\pi\epsilon_o(\sqrt{2})} \right]$$

$$W = \frac{1}{4\pi\epsilon_o} \left(Q_1 Q_2 + Q_1 Q_3 + \frac{Q_2 Q_3}{\sqrt{2}} \right) = 9 \left(\frac{12}{\sqrt{2}} - 7 \right) \text{ nJ} = 13.37 \text{ nJ}$$

as obtained previously.